



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Main)

UNIT TEST

24-12-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TTAS, TOAS, TNAS, TRAS, TMAS, TIAS, I(A) & LIVE-I(A)]

Time : 3 Hours

PAPER-2

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in JEE(Main) 2022

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SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. With which of the following can the intrinsic semiconductor silicon be doped in order to obtain p-type silicon semiconductor ?

- (A) boron
- (B) phosphorus
- (C) carbon
- (D) arsenic

2. X and Y in figure I and II represents two sets (shaded area). The output represented by the set, F, in figure III is realized by the logic gate

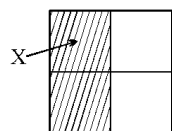


Figure I

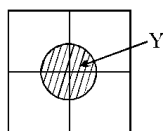


Figure II

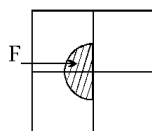


Figure III

- (A) OR
- (B) NOT
- (C) NOR
- (D) AND

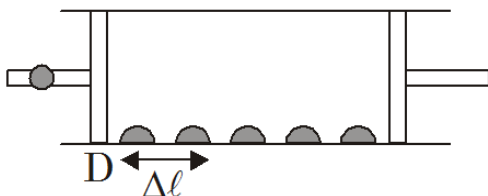
3. Which of the following statement is valid about the p-n junction ?

- (A) Electrons encounter no barrier at the junction in moving from the p region to the n region
- (B) Electrons encounter no barrier at the junction in moving from the n region to the p region
- (C) Holes encounter no barrier at the junction in moving from the p region to the n region
- (D) None of these

4. When light is incident from air to glass at an angle 57° , the reflected beam is completely polarized. If the same beam is incident from water to glass, the angle of incidence at which reflected beam is completely polarized will be

- (A) $\theta = 57^\circ$
- (B) $\theta > 57^\circ$
- (C) $\theta < 57^\circ$
- (D) cannot be determined

5. In a Kundt's tube distance between two consecutive heap is $\Delta \ell$ for air, while it is increased by 50% for a gas in the same tube with same resonate. If speed of sound in air is $\frac{1000}{3}$ m/s then speed of sound in gas at same temperature is



- (A) 1500 m/s
(B) 1000 m/s
(C) 500 m/s
(D) 250 m/s
6. A closed organ pipe of length 1.2m vibrates in its first overtone mode. The pressure variation is minimum at (neglect end correction)
- (A) 0.8m from the open end
(B) 0.4m from the open end
(C) Closed end
(D) 1.0m from the open end

7. The air column in a pipe closed at one end is made to vibrate in its first overtone by a tuning fork of frequency 550 Hz. The speed of sound in air is 330 m/s. End correction may be neglected. Let P_0 denotes the mean pressure at any point in the pipe and ΔP_0 the maximum amplitude of pressure variation in the pipe. Then choose the incorrect :-

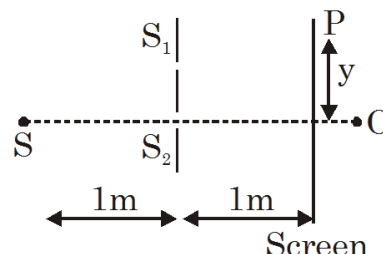
- (A) The length of the pipe is 45 cm.
(B) The length of the pipe is 75 cm.
(C) The maximum pressure at the middle of the pipe is $\left(P_0 + \frac{\Delta P_0}{\sqrt{2}} \right)$
(D) The maximum pressure at the closed end of the pipe is $(P_0 + \Delta P_0)$

8. Nuclei of radioactive element A are being produced at a constant rate α . The element has a decay constant λ . At time $t = 0$, there are N_0 nuclei of the element. Which of the following statements is correct ?

- (A) The number of nuclei of A at time t is $\frac{1}{2\lambda} (\alpha - (\alpha - \lambda N_0) e^{-\lambda t})$
(B) The number of nuclei of A at time t is $\frac{1}{\lambda} \{ (\alpha - \lambda N_0) e^{-\lambda t} \}$
(C) If $\alpha = 2N_0\lambda$, the number of nuclei of A after one half life of A is $\frac{3}{2}N_0$
(D) The number of nuclei of A after a very long time is $3N_0$

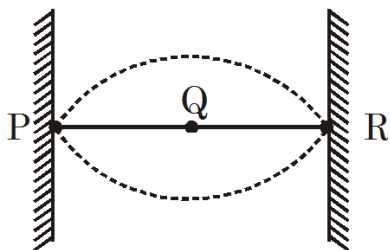
9. When a monochromatic light from a point source falls on a photo sensitive surface, the maximum kinetic energy of photoelectrons emitted is $K_{\max}$. If the distance between the source and the surface is doubled:
- (A) Number of photoelectrons coming out of the surface becomes double.
- (B) Number of photoelectrons coming out of the surface becomes one-fourth of the initial value.
- (C) $K_{\max}$ gets doubled
- (D) None
10. The pitch of a screw gauge is 1 mm and there are 100 divisions on its circular scale. When nothing is put in between its jaws, the zero of the circular scale lies 4 divisions below the reference line. When a steel wire is placed between the jaws, two main scale divisions are clearly visible and 67 divisions on the circular scale are observed. The diameter of the wire is
- (A) 2.71 mm
- (B) 2.67 mm
- (C) 2.63 mm
- (D) 2.65 mm

11. In an arrangement shown in the figure, $d \ll D$, where d is the distance of separation of the slits S_1 and S_2 and D is the distance between the slits and the screen. Source S is emitting monochromatic light of wavelength λ . Then



- (A) Minimum value of d for the dark fringe at O is $\sqrt{\frac{\lambda}{2}}$
- (B) Minimum value of d for the dark fringe at O is $\sqrt{\frac{\lambda}{3}}$
- (C) Minimum value of d for the dark fringe at O is $\sqrt{\frac{\lambda}{6}}$
- (D) Minimum value of d for the dark fringe at O is $\sqrt{\frac{\lambda}{8}}$

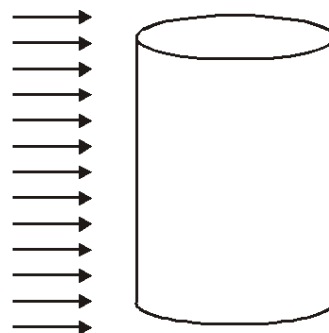
12. A string PQR is fixed between two rigid supports separated by distance L . The string is vibrating in its fundamental mode as shown in the figure. Choose the correct alternative. (Point Q is the mid-point of the string)



- (A) Energy of the portion PQ is constant.
 (B) Point Q on the string will have maximum potential energy when its kinetic energy is zero.
 (C) Energy of the string PQR is constant but transfer of energy takes place between portions PQ and QR.
 (D) None of these
13. A radio nuclide with half-life T days emits β -particle of average kinetic energy E joule. This radionuclide is used as source in a machine which generates electric energy with efficiency η . The number of moles of the nuclide required to generate electric energy at an initial rate P is : (Avogadro number = N)

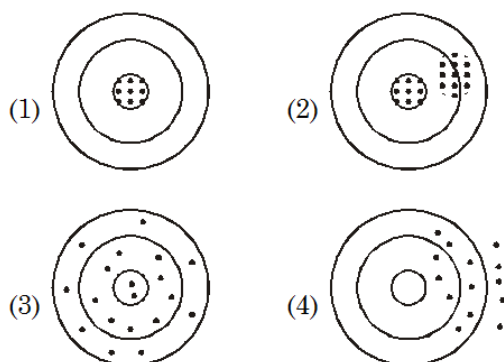
- (A) $\frac{PT \ln 2}{\eta NE}$ (B) $\frac{PT}{\eta NE}$
 (C) $\frac{PT}{\eta NE \ln 2}$ (D) None of these

14. Find the force exerted by a light beam of intensity I , incident on a cylinder (height h and base radius R) placed on a smooth surface as shown in figure if surface of cylinder is perfectly reflecting. (Speed of light is c)



- (A) $\frac{8IRh}{3c}$
 (B) $\frac{2IRh}{3c}$
 (C) $\frac{4IRh}{3c}$
 (D) None
15. An object of specific gravity ρ is hung from a thin steel wire. The fundamental frequency for transverse standing waves in the wire is 300 Hz. The object is immersed in water so that one half of its volume is submerged. The new fundamental frequency in Hz is
- (A) $300 \left(\frac{2\rho - 1}{2\rho} \right)^{1/2}$
 (B) $300 \left(\frac{2\rho}{2\rho - 1} \right)^{1/2}$
 (C) $300 \left(\frac{2\rho}{2\rho - 1} \right)$
 (D) $300 \left(\frac{2\rho - 1}{2\rho} \right)$

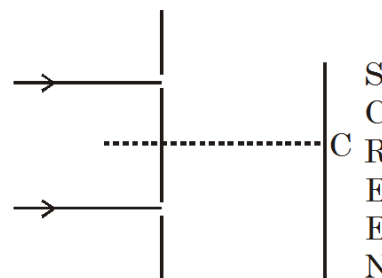
16. A demonstration was conducted to understand systematic and random errors in physics experiments via an analogy. In this analogy, conduction of physics experiment is compared to throwing of darts towards a target. A stationary gun is mounted and is used to throw darts towards a target held by a marksman. The gun should be adjusted so that the darts hit exactly the center of the target. Gun hitting the center of target is compared to accurate measurement taken by a person doing physics experiment. Just like there can be random errors in physics experiment, similarly there can be random errors in this experiment like shaking of marksman, random and turbulent wind present in region between target and gun. Also there can be systematic errors. The demonstration is done four times under different conditions and the points at which dart hits the target in shown in the following four figures.



If we define systematic errors and random errors in physics experiments and this demonstration in similar way, then choose the correct statement.

- (A) Demonstration 1 has low systematic error and high random error.
 (B) Demonstration 2 has low systematic error and high random error.
 (C) Demonstration 3 has high systematic error and high random error.
 (D) None

17. If white light is used in a Young's double – slit experiment & point C represents centre of a screen then choose the incorrect option.



- (A) Bright white fringe is formed at the centre of the screen
 (B) Fringe closest to either side of centre of the screen is red
 (C) Fringe closest to either side of centre of the screen is violet
 (D) After few fringes no clear fringe pattern is seen
18. If the plank's constant would be double the present value, in the Bohr's model for hydrogen atom then choose the incorrect statement.
- (A) The speed of the electron in the ground state would be halved
 (B) The radius of electronic orbit in the ground state would be four times
 (C) The energy required to ionize the hydrogen atom in the ground state would be doubled
 (D) The angular momentum of the electron in the ground state would be doubled

19. In a hypothetical atom, a negatively charged particle having a charge of magnitude $3e$ and mass $3m$ revolves around a proton. Here, e is the electronic charge and m is the electronic mass. Mass of proton may be assumed to be much larger than that of the negatively charged particle, thus the proton is at rest. This "atom" obeys Bohr's postulate of quantization of angular momentum, that is $mvr = n \left(\frac{h}{2\pi} \right)$. It is given that for the first Bohr orbit of hydrogen atom: radius of orbit is r_0 , speed of electron is V_0 , and total energy is $-E_0$. Radius of hypothetical atom is

(A) r_0 (B) $\frac{r_0}{9}$ (C) $\frac{r_0}{27}$ (D) $\frac{r_0}{81}$

20. In a hypothetical atom, a negatively charged particle having a charge of magnitude $3e$ and mass $3m$ revolves around a proton. Here, e is the electronic charge and m is the electronic mass. Mass of proton may be assumed to be much larger than that of the negatively charged particle, thus the proton is at rest. This "atom" obeys Bohr's postulate of quantization of angular momentum, that is $mvr = n \left(\frac{h}{2\pi} \right)$. It is given that for the first Bohr orbit of hydrogen atom: radius of orbit is r_0 , speed of electron is V_0 , and total energy is $-E_0$. The momentum of an emitted photon when it makes a transition from the second excited state to ground state, is

(A) $\frac{8E_0}{c}$ (B) $\frac{E_0}{c}$
(C) $\frac{24E_0}{c}$ (D) None of these

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

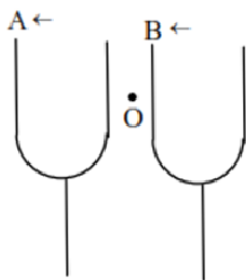
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. Consider the nuclear reaction ${}^{81}_{36}\text{Kr} + {}^0_{-1}\text{e} \rightarrow {}^{81}_{35}\text{Br} + {}^0_0\nu$. The mass of Kr is 80.9166 amu, mass of Br is 80.9163 amu, mass of electron is 0.00054858 amu and neutrino can be assumed to be massless. If it behaves like a photon, Br will recoil after the nuclear reaction. What is the recoil velocity of Br (in km/s) ?
Take 1 amu = 931 MeV/ C^2 .

2. Two tuning forks A and B each of natural frequency 85 Hz move with velocity 10 m/s relative to stationary observer 'O'. Fork A moves away from the observer while the fork B moves towards him as shown in the figure. A wind is blowing with a speed 10 m/s in the direction of motion of fork A. Find the beat frequency measured by the observer in Hz. [Take speed of sound in air as 340 m/s]

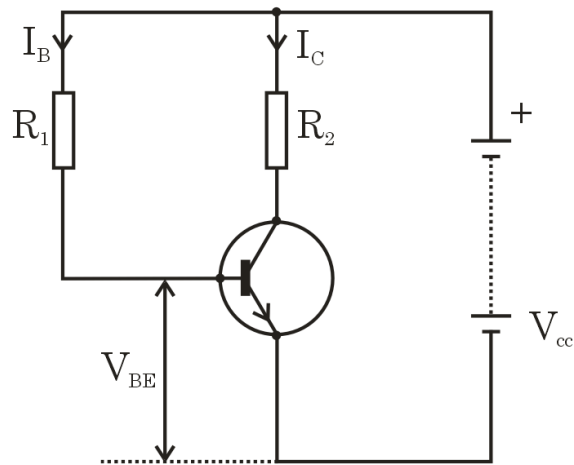


3. Two radio active nuclei A and B decays into C with decay constant λ and 2λ respectively. Initially both A and B have same number of nuclei N_0 . The rate of production of C after time $\frac{\ln 2}{\lambda}$ is $\alpha\lambda N_0$. Find α .
4. The peak emission from a black body at a certain temperature occurs at a wavelength of 6000 Å. On increasing its temperature, the total radiation emitted is increased 16 times. These radiations are allowed to fall on a metal surface. Photoelectrons emitted by the peak radiation at higher temperature can be brought to rest by applying a potential equivalent to the excitation potential corresponding to the transition for the level $n = 4$ to $n = 2$ in the Bohr's hydrogen atom. The work function of the metal is given by $\frac{\alpha}{100}$ eV where α is the numerical constant. The value of α is found $53 \times x$, find x . [Take: $hc = 12420 \text{ eV-Å}$]

5. A neutron strikes a hydrogen atom which is initially at rest and free to move. The initial speed of the neutron is v_0 . After the collision both neutron and the H atom move making 30° angle with the initial velocity direction of the neutron. Also, the H atom is found to get excited to first excited state. If the initial KE of neutron is 'x' times first excitation energy of the H atom, find the value of 'x' (assume mass of neutron = mass of H atom)
6. A free neutron at rest, decays into three particles: a proton, an electron and an antineutrino. ${}_0^1n \rightarrow {}_1^1p + {}_1^0e + \bar{\nu}$ The rest masses are :
- $$m_n = 939.5656 \text{ MeV}/c^2,$$
- $$m_p = 938.2723 \text{ MeV}/c^2,$$
- $$m_e = 0.5109 \text{ MeV}/c^2.$$
- In a particular decay, the antineutrino was found to have a total energy (including rest mass energy) of 0.0004 MeV and the momentum of proton was found to be equal to the momentum of electron. Find the kinetic energy of the electron (in MeV).

7. Two quantities A and C are measured independently. C has dimensions same as that of A^2 . Values of A and C measured in SI unit is
 $A = 20 \pm 0.1$
 $C = 300 \pm 2$
 Another quantity x is defined as $x = A^3 - AC$ and is calculated using the measured values of A and C. If the percentage error in the calculation of X is $1.3n\%$, find 'n'
8. The relative density of a ball can be found by hanging the ball from a spring balance and noting the reading. The balance reads $4.00 \pm 0.02\text{N}$ in air and $1.00 \pm 0.02\text{N}$ in water. The maximum percentage error in measurement of relative density is $\frac{P}{6}$. Find the value of 'P' to the nearest integer.
9. A sound source emits frequency of 175 Hz when moving towards a rigid wall with speed 5 m/s and observer is moving away from wall with same speed 5 m/s. Both source and observer moves on a straight line which is perpendicular to the wall. The number of beats per second heard by the observer will be [Speed of sound = 355 m/s] Source is in between observer and wall.

10. In the junction transistor voltage amplifier circuit of figure, if $R_1 = 100\text{ k}\Omega$, $R_2 = 1\text{ k}\Omega$, $V_{cc} = 6.0\text{ V}$ and $V_{BE} = 1\text{ V}$, current gain = 80 then voltage across R_2 is (in V)



PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

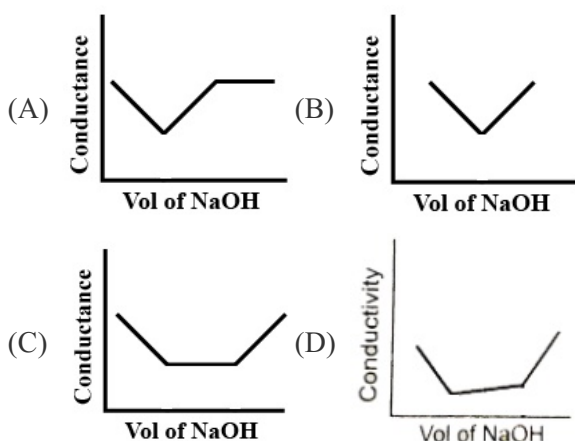
Negative Marks : -1 If wrong option is selected.

- Knowing that the chemistry of lanthanoids (Ln) is dominated by its + 3 oxidation state, which of the following statements is incorrect?
 (A) Ln(III) compounds are generally colourless
 (B) Ln(III) hydroxides are mainly basic in character
 (C) Because of the large size of the Ln(III) ions the bonding in its compounds is predominantly ionic in character
 (D) The ionic sizes of Ln(III) decrease in general with increasing atomic number
- Element "X" shows diagonal similarity with element "Y". Oxide of "X" in fused condition when mixed with Na_2CO_3 produces many different types of compounds of "X". The fluoride of "Y" forms an adduct with NH_3 . Which is incorrect for "X" & "Y" or their compounds.
 (A) $(\text{X}_4\text{O}_{11})_n^{6n-}$ has at least two oxygen shared per tetrahedral unit.
 (B) H_3YO_3 is a monobasic lewis acid.
 (C) XO_2 has 3-D network structure.
 (D) YN(s) has structure identical to Benzene & is called Inorganic Benzene

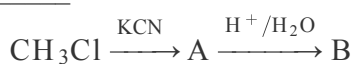
- Select the incorrect statements from the following options-
 (A) In the Haber's process of NH_3 formation, the catalyst used is oxide of Fe with small amount of K_2O and Al_2O_3
 (B) In the deacon process the catalyst used is cupric chloride
 (C) White phosphorus is insoluble in water but soluble in CS_2
 (D) N-O bond attached to H-atom in HNO_3 is shorter than other two N-O bonds.
- Which of the following exists and is stable ?
 (A) FeI_3 (B) MnF_7
 (C) CuF (D) VF_5
- Which of the following ions in aqueous solution is green coloured ?
 (A) $\text{Fe}^{2+}(\text{aq})$ (B) $\text{Cu}^{2+}(\text{aq})$
 (C) $\text{Zn}^{2+}(\text{aq})$ (D) $\text{Mn}^{2+}(\text{aq})$
- Which of the following causes laxative effect when present above 500 ppm in water :
 (A) NO_3^- (B) SO_4^{--}
 (C) F^- (D) Zn^{++}
- When one mole of $\text{Hg}_2(\text{NO}_3)_2$ is added to 1dm^3 of a solution containing 3 mole of NaCl then the concentration of Hg_2^{2+} in final solution is $[\text{Ksp}(\text{Hg}_2\text{Cl}_2) = 10^{-18}]$.
 (A) 10^{-6} M (B) 10^{-12} M
 (C) 10^{-15} M (D) 10^{-18} M

8. The self ionization constant for pure formic acid $K_{\text{sic}} = [\text{HCOOH}_2^+][\text{HCOO}^-]$ has been estimated as 10^{-6} M^2 at room temperature. The density of formic acid is 1.15 g/cm^3 . What percentage of formic acid molecule is converted to formate anion:
- (A) 0.001 %
 (B) 0.002%
 (C) 0.003%
 (D) 0.004%
9. A 0.010 M solution of $\text{ZrO}(\text{NO}_3)_2$ was found to have a pH of 3.80 [Antilog $(-3.8) = 1.6 \times 10^{-4}$]. What is the value of K_b for $\text{ZrO}(\text{OH})^+$ at 25°C ?
- (A) 1.0×10^{-6}
 (B) 3.9×10^{-6}
 (C) 3.9×10^{-9}
 (D) 3.9×10^{-10}
10. Which of the following statements is not true for a lyophilic sol
- (A) It is heavily solvated
 (B) Strong interactions exist between dispersed phase and dispersion medium
 (C) The coagulation of this sol is reversible in nature
 (D) It is not stable
11. Identify correct statements-
- (i) If on adding electrolyte in an emulsion the conductivity increases significantly, then it will be oil in water type emulsion.
 (ii) Formation of micelles take place only above a particular temperature called Kraft temperature.
 (iii) Zeolites are widely used in petrochemical industries for cracking of hydrocarbon and their isomerisation.
 (iv) Physical adsorption is usually multilayered.
- (A) i, ii, iii only
 (B) i, iii only
 (C) i, ii, iii, iv
 (D) i, iii, iv only
12. If $E_{\text{Tl}^{+1}/\text{Tl}}^\circ = -0.336\text{V}$, $E_{\text{Tl}^{+3}/\text{Tl}^{+1}}^\circ = 1.25 \text{ V}$ then the value of $E_{\text{Tl}^{+3}/\text{Tl}}^\circ$ is -
- (A) 0.814 V (B) 0.721 V
 (C) 1.325 V (D) 1.721 V
13. Determine λ_{eq}^0 for Mohr salt " $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ " (Neglect the effect of hydrated water molecules)
 Given : $\lambda_m^0(\text{Fe}^{2+}) = a$; $\lambda_m^0(\text{NH}_4^+) = b$; $\lambda_m^0(\text{SO}_4^{2-}) = c$
 (All quantities are in SI units)
- (A) $\frac{a}{2} + b + \frac{c}{2}$
 (B) $2a + b + 2c$
 (C) $4(a + 2b + 2c)$
 (D) $\frac{a + 2b + 2c}{4}$

14. Conductometric titration curve of equimolar mixture of a HCl and HCN with NaOH(aq) is:



15. In the following reactions, the final product B is _____



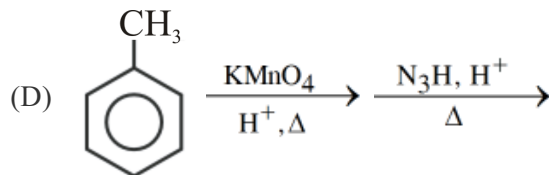
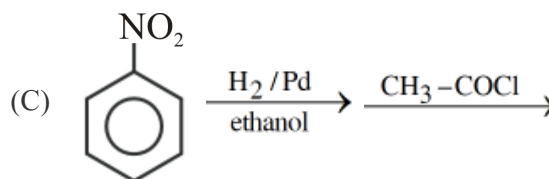
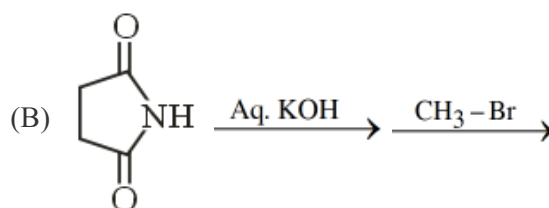
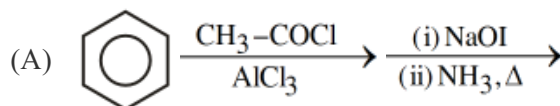
- (A) CH_3COOH (B) HCOOH
(C) CH_3NH_2 (D) CH_3COCH_3
16. A compound 'X' on reaction with ethanol gives sweet smelling compound. 'X' on reaction with $\text{Cl}_2/\text{Red P}$ and then on treatment with water yields 'Y'. 'Y' reacts with KCN to form 'Z' which on hydrolysis and then heating yields 'X' back along with an acidic colourless gas. Which of the following can be the compound 'X' ?

- (A) $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3$
(B) Me_3CCOOH
(C) CH_3COOH
(D)

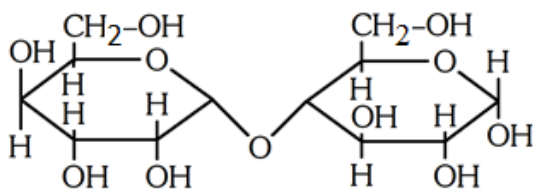
17. Which of the following compounds will not yield amine on reaction with Br_2/NaOH ?

- (A) Ethanamide
(B) Benzamide
(C) N-methyl ethanamide
(D) All of these

18. Which of the following reaction product will give positive isocyanide test ?

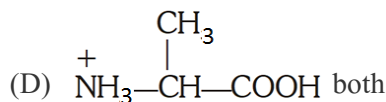
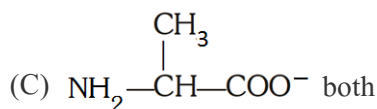
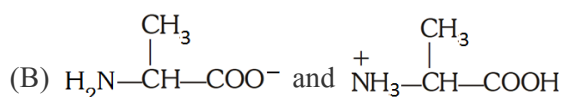
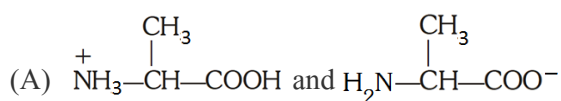


19. Incorrect statement about given carbohydrate is



- (A) Above compound is a reducing sugar
 (B) Above compound show mutarotation
 (C) Above compound is a non-reducing sugar
 (D) Above compound has a glycosidic linkage

20. Structure of alanine at pH = 2 and pH = 10 are respectively.



SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

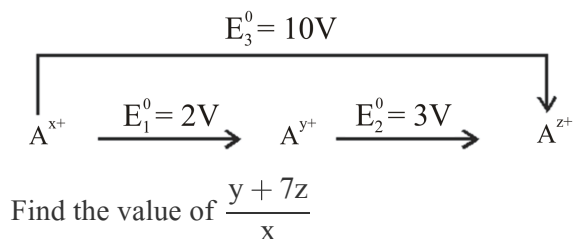
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

- Thiosulphate ion is oxidised by KMnO_4 in slightly alkaline medium. Find sum of oxidation number of Mn and sulphur in products.
- Number of non-radioactive alkali metals forming superoxide as major product on heating with excess $\text{O}_2 = x$,
 Number of II-A metals of periodic table whose hydrated halides suffer hydrolysis on heating = y.
 Number of non radio active alkali metals which dissolve in liquid ammonia and produce blue colour solution = u
 The value of $(x + y - u)$ would be....

3. For the given Latimer diagram (E^0 = standard half cell potential)
(such that $x > y > z$)



4. On addition of 10 ml solution of 10% NaCl to 100 ml gold sol in the presence of 0.0250 gm of a lyophilic sol the coagulation is just prevented. Find gold number of the lyophilic sol?

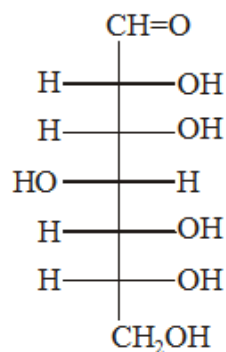
5. A hydrogen electrode placed in two different buffer solutions, say 'A' and 'B', both containing CH_3COONa and CH_3COOH . If molar ratios of $\text{CH}_3\text{COONa} : \text{CH}_3\text{COOH}$ in A and B are $x : y$ and $y : x$ respectively, then pK_a value of CH_3COOH at 25°C will be :-

Given that for same hydrogen electrode, oxidation potentials in buffer A and B are 0.36 volt and 0.12 volt respectively.

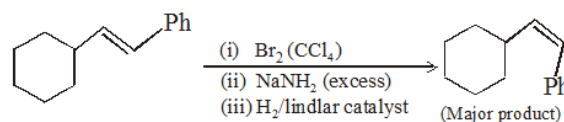
[take $\frac{2.303RT}{F} = 0.06$]

6. How many following statement(s) are correct ?

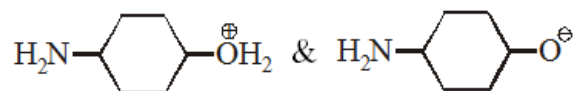
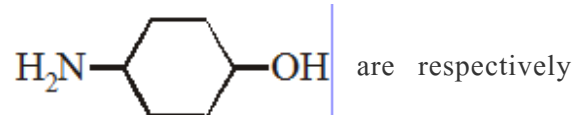
- (i) Bakelite is thermosetting polymer.
(ii) This can be the fischer projection of D-aldoheptose



- (iii)

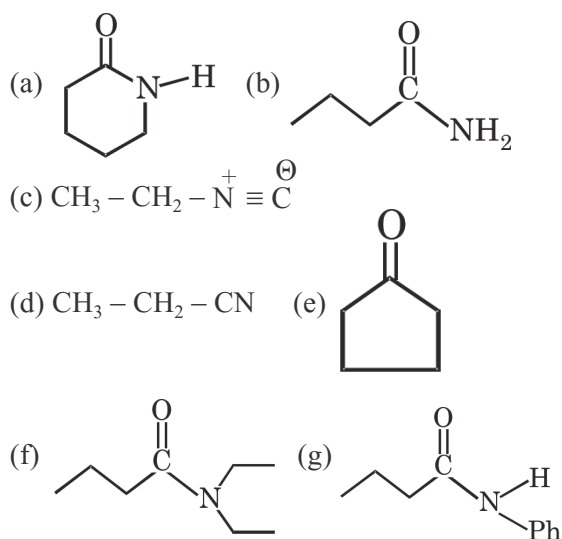


- (iv) The structure of conjugate acid & conjugate base of the following compound



- (v) $\text{Bromocyclopentene} \xrightarrow{\text{Zn}}$ In this reaction product formed have aromatic character

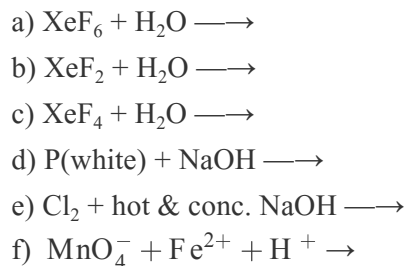
7. How many of the following compounds form 2° -amines on reaction with LiAlH_4



8. Glucose $\xrightarrow[\text{(iv) NO}_2\text{BF}_4]{\text{(i) P + HI / } \Delta}$ 'X' (Major)
- (ii) $\text{Al}_2\text{O}_3 + \text{Cr}_2\text{O}_3 / \Delta$
 (iii) $\text{CO} + \text{HCl} / \text{AlCl}_3$

The position of locant of substituent group in major product (X) is?

9. How many reactions are example of disproportionation reaction?



10. How many of the following are paramagnetic in nature.

Fe_2O_3 , MnO , La_2O_3 , ZnO , AgCl , SO_3 , PCl_5 ,
 CO , NO , PmO , Eu_2O_3 , U_2O_3 , MnO_2

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- Let v_1 = variance of $\{13, 16, 19, \dots, 103\}$ and v_2 = variance of $\{20, 26, 32, \dots, 200\}$, then $v_1 : v_2$ is
(A) 1 : 2 (B) 1 : 1
(C) 4 : 9 (D) 1 : 4
- Statement $(p \rightarrow q) \leftrightarrow (q \vee \sim p)$ is
(A) Equivalent to $p \wedge q$
(B) Tautology
(C) Fallacy
(D) Neither tautology nor fallacy
- Let X be a set containing n elements. If two subsets A and B of X are picked at random, the probability that A and B have the same number of elements, is
(A) $\frac{2^n C_n}{2^n}$
(B) $\frac{1}{2^n C_n}$
(C) $\frac{1.3 \dots (2n-1)}{2^n \cdot n!}$
(D) $\frac{3^n}{4^n}$

- If a drunkard person tries to take a step, then it will be a forward or backward step with probabilities $\frac{1}{4}, \frac{1}{2}$ respectively, or he will remain in 'as it is' position. If he tries to take a step 5 times, then probability that he will be one step away from the initial position, is
(A) $\frac{210}{2^8}$ (B) $\frac{315}{2^{10}}$
(C) $\frac{171}{2^{16}}$ (D) $\frac{75}{2^8}$
- Let C be the set of complex numbers, $z \in C$ satisfies, $|z-3| + |z+1| \geq 8$, then minimum value of $|z-1|$ is-
(A) $\sqrt{2}$ (B) $2\sqrt{2}$
(C) 3 (D) $2\sqrt{3}$
- Area bounded by the curve, $-8z^2 - 8\bar{z}^2 + 20z\bar{z} = 144$ is (where z is a complex number)-
(A) π (B) 6π
(C) 12π (D) 4π
- Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that $5\vec{a} - 3\vec{b}$ and $3\vec{a} + 4\vec{b}$ are perpendicular to each other, then the value of $|\vec{a} \times \vec{b}|$ is -
(A) $\frac{3}{11}$ (B) $\frac{\sqrt{112}}{11}$
(C) $\frac{4}{11}$ (D) $\frac{\sqrt{105}}{11}$

8. Let $\vec{v}$ be a unit vector which follows the equation, $\vec{v} \times \vec{b} = \vec{c}$. Also, $|\vec{b}| = 2$ and $|\vec{c}| = \sqrt{3}$ then
- (A) $\vec{v} = -\vec{b} + \vec{b} \times \vec{c}$
- (B) $\vec{v} = \frac{3}{4}(\vec{b} + 2\vec{b} \times \vec{c})$
- (C) $\vec{v} = \frac{1}{4}(\vec{b} + \vec{b} \times \vec{c})$
- (D) $\vec{v} = \frac{\vec{b} \times \vec{c}}{4}$
9. A variable plane passes through fixed point (1,-2,3) and meets the coordinate axes at A,B,C. The point of intersection of planes through A,B,C parallel to coordinate planes lies on -
- (A) $xy - \frac{1}{2}yz + \frac{1}{3}zx = 6$
- (B) $yz - 2zx + 3xy = xyz$
- (C) $xy - 2yz + 3zx = 3xyz$
- (D) $-xy + \frac{1}{2}yz - \frac{1}{3}zx = 6$
10. An insurance agent insured 50 scooters, 40 cars and 60 trucks in a month. Statistics show that respective probabilities of accidents of these vehicles are 0.02, 0.06 and 0.15. One of his insured vehicles applied for a claim. Probability that the vehicle was a scooter, is -
- (A) $\frac{5}{62}$ (B) $\frac{3}{62}$
- (C) $\frac{1}{55}$ (D) $\frac{1}{56}$
11. Let $\lambda = 2(1+w)(1+w^2) + 3(2+w)(2+w^2) + 4(3+w)(3+w^2) + \dots + 21(20+w)(20+w^2)$, then number of zeroes at the end of λ is (where w is imaginary cube root of unity) -
- (A) 0 (B) 1
- (C) 2 (D) 3
12. Let $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $|\vec{c}| = 5$, then the value of $|\vec{a} + \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} + \vec{a}|^2$ is equal to :
- (A) 104 (B) 114
- (C) 124 (D) 134
13. Let $P_1 = x + y + z + 1 = 0$, $P_2 = x - y + 2z + 1 = 0$, $P_3 = 3x + y + 4z + 7 = 0$ be three planes. Find the distance of line of intersection of planes $P_1 = 0$ and $P_2 = 0$ from the plane $P_3 = 0$.
- (A) $\frac{2}{\sqrt{26}}$ (B) $\frac{4}{\sqrt{26}}$
- (C) $\sqrt{\frac{1}{26}}$ (D) $\frac{7}{\sqrt{26}}$
14. Mean square deviation of set of observations $x_1, x_2, x_3, x_4, \dots, x_n$ about a point ℓ is defined as $\frac{1}{n} \sum_{i=1}^n (x_i - \ell)^2$, if mean square deviation about -1 and 1 of a set of observations are 7 and 3 respectively. Then standard deviation of these observations about the mean is-
- (A) $\sqrt{2}$ (B) $\sqrt{3}$
- (C) 2 (D) 5

15. Statement "If $x^2 - 5x + 6 = 0$, then $x = 3$ OR $x = -3$ " is equivalent to

(A) If $x^2 - 5x + 6 \neq 0$, then $x \neq 3$ and $x \neq -3$

(B) If $x^2 - 5x + 6 = 0$, then $x = 3$ and $x = -3$

(C) If $x \neq 3$, then $x \neq -3$ and $x^2 - 5x + 6 = 0$

(D) If $x \neq 3$, then $x \neq -3$ and $x^2 - 5x + 6 \neq 0$

16. Probability that the product of the outcomes when three dice are rolled simultaneously is divisible by 4 is equal to

(A) $\frac{3}{8}$ (B) $\frac{1}{2}$

(C) $\frac{5}{8}$ (D) $\frac{3}{4}$

17. Mr. A randomly picks 3 distinct number from set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and arrange them in descending order to get a 3 digit number. Mr. B randomly picks 3 distinct number from set $\{1, 2, 3, 4, 5, 6, 7, 8\}$ also arranges them in descending order to get 3 digit number. Find probability that Mr A's 3 digit number is greater than Mr B's 3 digit number

(A) $\frac{2}{3}$ (B) $\frac{37}{56}$

(C) $\frac{139}{168}$ (D) $\frac{39}{59}$

18. A plane passing through $(1, 1, 1)$ cut positive direction of co-ordinate axes at A, B and C, then the volume of tetrahedron OABC (as V) satisfies

(A) $V \leq \frac{9}{2}$ (B) $V \geq \frac{9}{2}$

(C) $V = \frac{7}{2}$ (D) None

19. Given that $P(A) = 2/7$, $P(B) = 2/7$,

$P(A \cap \bar{B} \cap C) = 1/14$,

$P(A \cap \bar{B} \cap \bar{C}) = 1/14$.

$P(\bar{A}|\bar{B}) = \frac{x}{y}$ equals

(A) $\frac{1}{2}$ (B) $\frac{4}{5}$

(C) $\frac{2}{7}$ (D) $\frac{4}{7}$

20. Let z_1, z_2, z_3 be roots of the equation

$z^3 + z^2(i - 1) + z(2 - i) - 2 = 0$, then

$\text{Re}(z_1) + \text{Re}(z_2) + \text{Re}(z_3)$ is -

(A) -1 (B) 0

(C) 1 (D) 2

SECTION-II : (Maximum Marks: 20)

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The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

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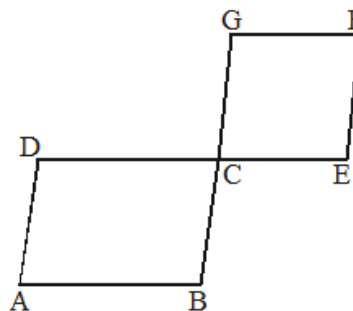
1. If nine numbers 1, 2, 3, 4, 5, 6, 7, 8, 9 are put into a matrix of order 3×3 randomly so that each number occur exactly once. If the probability that the sum of the numbers in atleast one row is greater than 21 is $\frac{1}{p}$, then the value of p is
2. z is a complex number satisfying the condition $\left| z - \frac{5}{z} \right| = 4$. Sum of greatest & least value of $|z|$ is

3. Consider $\vec{r} = a\hat{i} + b\hat{j} + c\hat{k}$. If α, β and γ be the projection of $\vec{r}$ along $\frac{2\hat{i} + \hat{j}}{\sqrt{5}}, \frac{-2\hat{i} + \hat{j}}{\sqrt{5}}$ and $\hat{k}$ respectively such that $|\alpha| = |\beta| = |\gamma|$ and $|\vec{r}| = 1$, then number of such ordered triplets (a, b, c) is
4. Let shortest distance between two opposite edges of a tetrahedron be '4 unit' and the length of these opposite edges are same and equal to 6 unit. If angle between these two opposite edges is 30° and volume of tetrahedron is V, the value of $\frac{V}{6}$ is
5. The plane $2x - 2y + z = 3$ is rotated about the line where it cuts the xy plane by an acute angle α . If the new position of plane contains the point (3, 1, 1) then $9 \cos \alpha$ equal to _____
6. Absolute value of a for which the lines $L_1 : \frac{x-2}{1} = \frac{y-9}{2} = \frac{z-13}{3}$ and $L_2 : \frac{x-a}{1} = \frac{y-7}{-2} = \frac{z+2}{3}$ intersect, is
7. Number of complex numbers z satisfying the equation $z^3 = \bar{z}$ & $\arg(z + 1) = \frac{\pi}{4}$ simultaneously is-

8. A parallelogram is constructed on $5\vec{a} + 2\vec{b}$ and $\vec{a} - 3\vec{b}$ where $|\vec{a}| = 2\sqrt{2}$ and $|\vec{b}| = 3$ & angle between $\vec{a}$ & $\vec{b}$ is $\frac{\pi}{4}$. If the magnitude of longer diagonal is $\sqrt{\alpha\beta r}$ where $\alpha\beta r$ is three digit number then $\alpha + \beta - r$ is equal to

9. z_1, z_2 are the roots of quadratic equation $z^2 + az + b = 0$ (a, b are complex).
If $|z_1 + z_2| = |z_1| + |z_2|$ then there exist positive real number λ such that $a^2 = \lambda b$. Find the minimum value of λ .

10. In the given figure, AB, DE and GF are parallel to each other and AD, BG and EF are parallel to each other. If $CD : CE = CG : CB = 2 : 1$,
then $\frac{\text{area of } (\triangle AEG)}{\text{area of } (\triangle ABD)}$ is equal to



Space for Rough Work